

E9: 309 ADL 14-12-2020

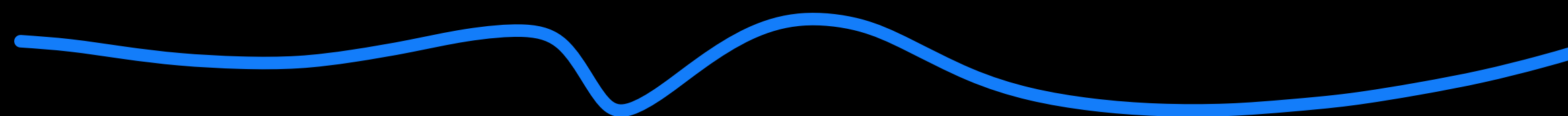
<http://leap.ee.iisc.ac.in/sriram/teaching/ADL2020/>

Housekeeping

Midterm project II - Abstract submission deadline 15/12/2020



Presentation deadline - Dec. 29th, 30th (time will be announced)



Recap from previous lecture

✳ Analyzing trained neural networks

- ✓ Hierarchical representations

Maximizing activations

Visualizing Higher-Layer Features of a Deep Network

Dumitru Erhan, Yoshua Bengio, Aaron Courville, and Pascal Vincent

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`first.last@umontreal.ca`

Technical Report 1341

Département d'Informatique et Recherche Opérationnelle



Learning the input pattern of a trained network

- ✳ Choose a trained neural network
- ✳ Find input patterns that maximize the activations from that neuron
- ✳ Solved using gradient ascent



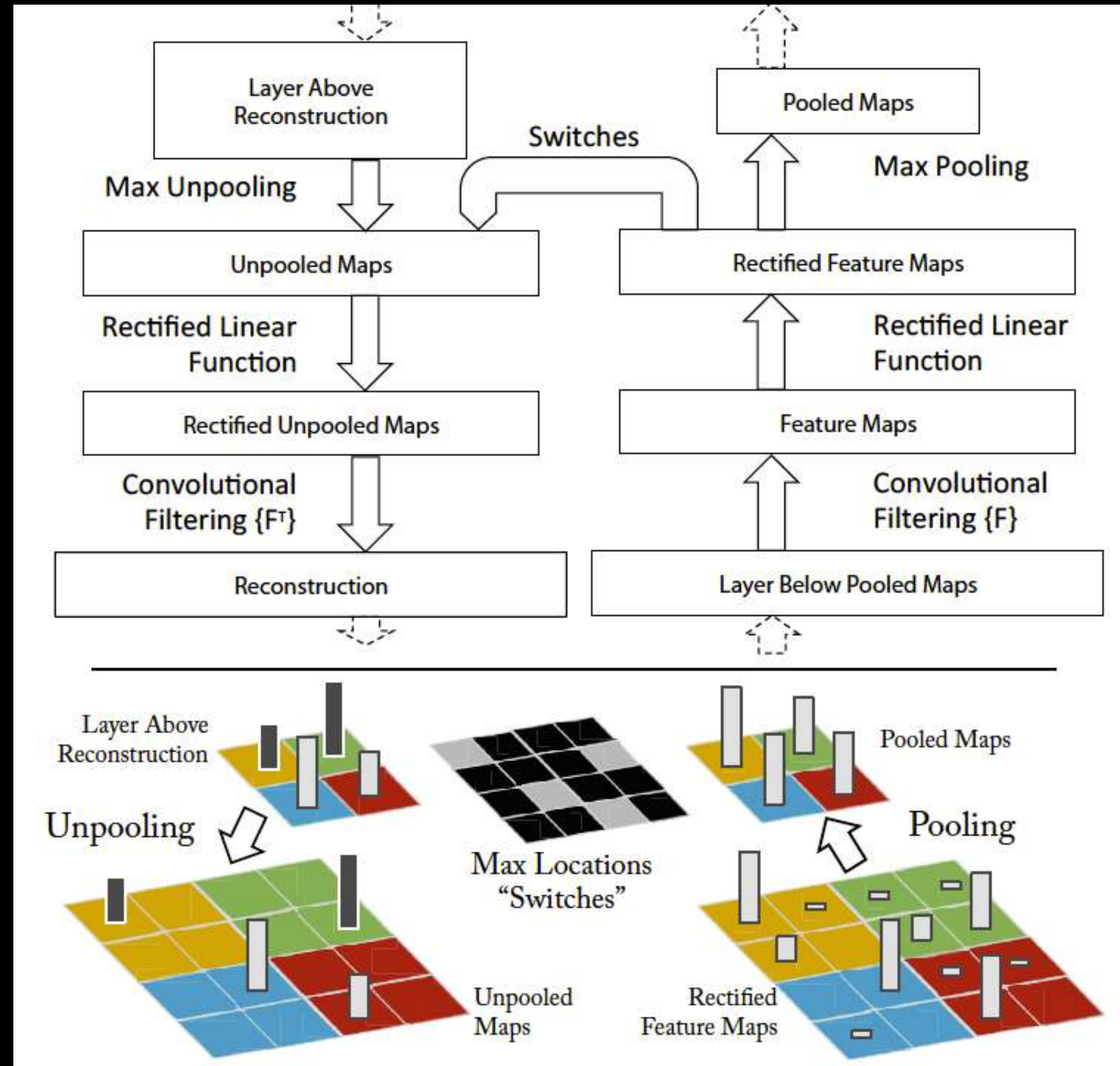
Hierarchical representations in deep networks

Visualizing and Understanding Convolutional Networks

Matthew D. Zeiler and Rob Fergus

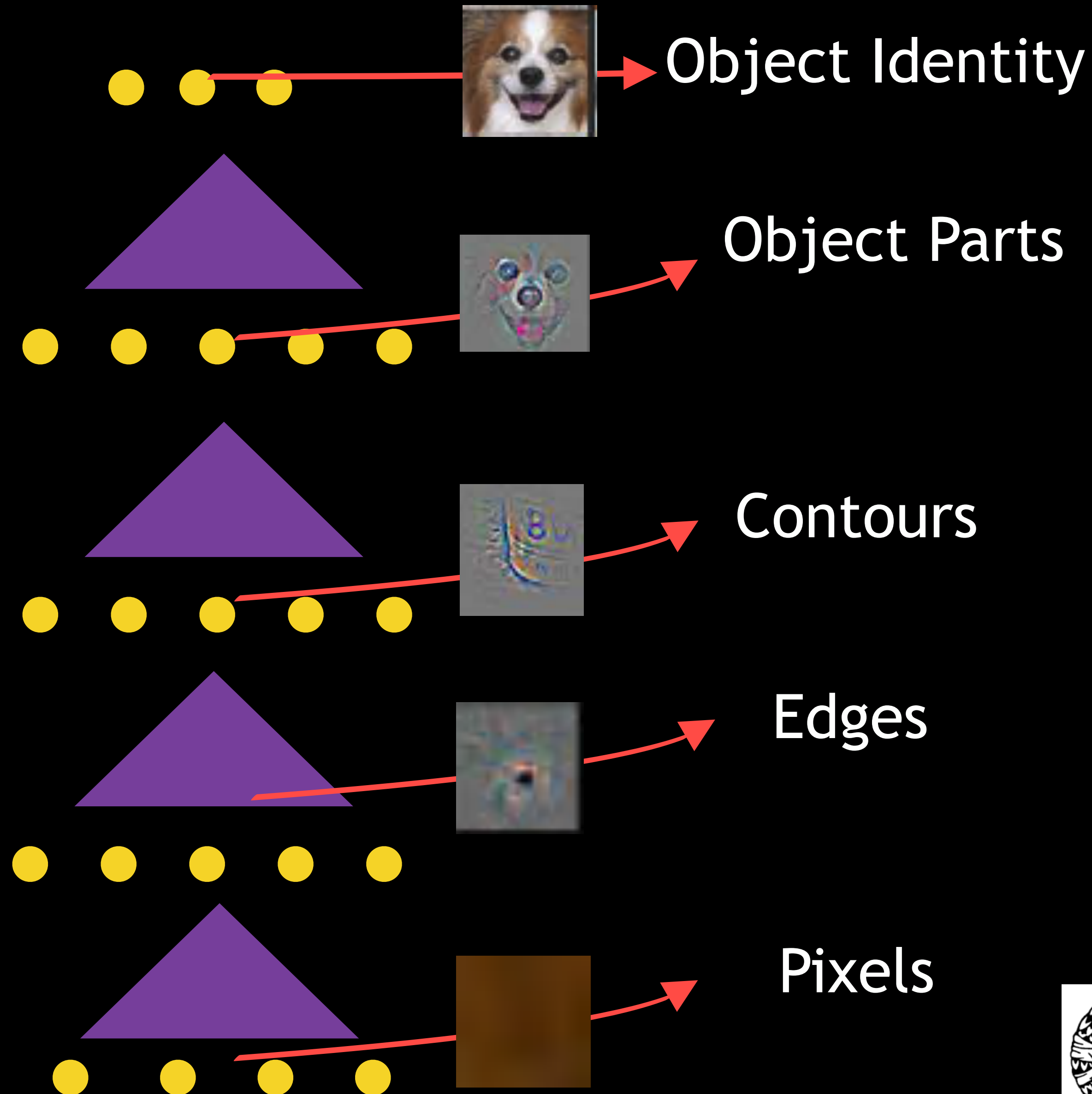
Dept. of Computer Science,
New York University, USA
{zeiler,fergus}@cs.nyu.edu

Hierarchical representations in deep networks



Hierarchical representations in deep networks

[Zeiler, 2014]



UNDERSTANDING HOW DEEP BELIEF NETWORKS PERFORM ACOUSTIC MODELLING

Garcia-Romero, Daniel, et al. "Speaker diarization using deep neural network embeddings." *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2017.

Abdel-rahman Mohamed, Geoffrey Hinton, and Gerald Penn

Department of Computer Science, University of Toronto

2012

t-SNE embeddings for visualization

IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS VOL. 23, NO. 1, JANUARY 2017

Visualizing the Hidden Activity of Artificial Neural Networks

Paulo E. Rauber, Samuel G. Fadel, Alexandre X. Falcão, and Alexandru C. Telea



t-SNE embeddings for visualization

SVHN dataset



t-SNE embeddings for visualization

MNIST dataset



t-SNE embeddings for visualization

CIFAR10 dataset

CIFAR 10



Understanding Deep Networks

tSNE
projection
of last layer
of the neural network.

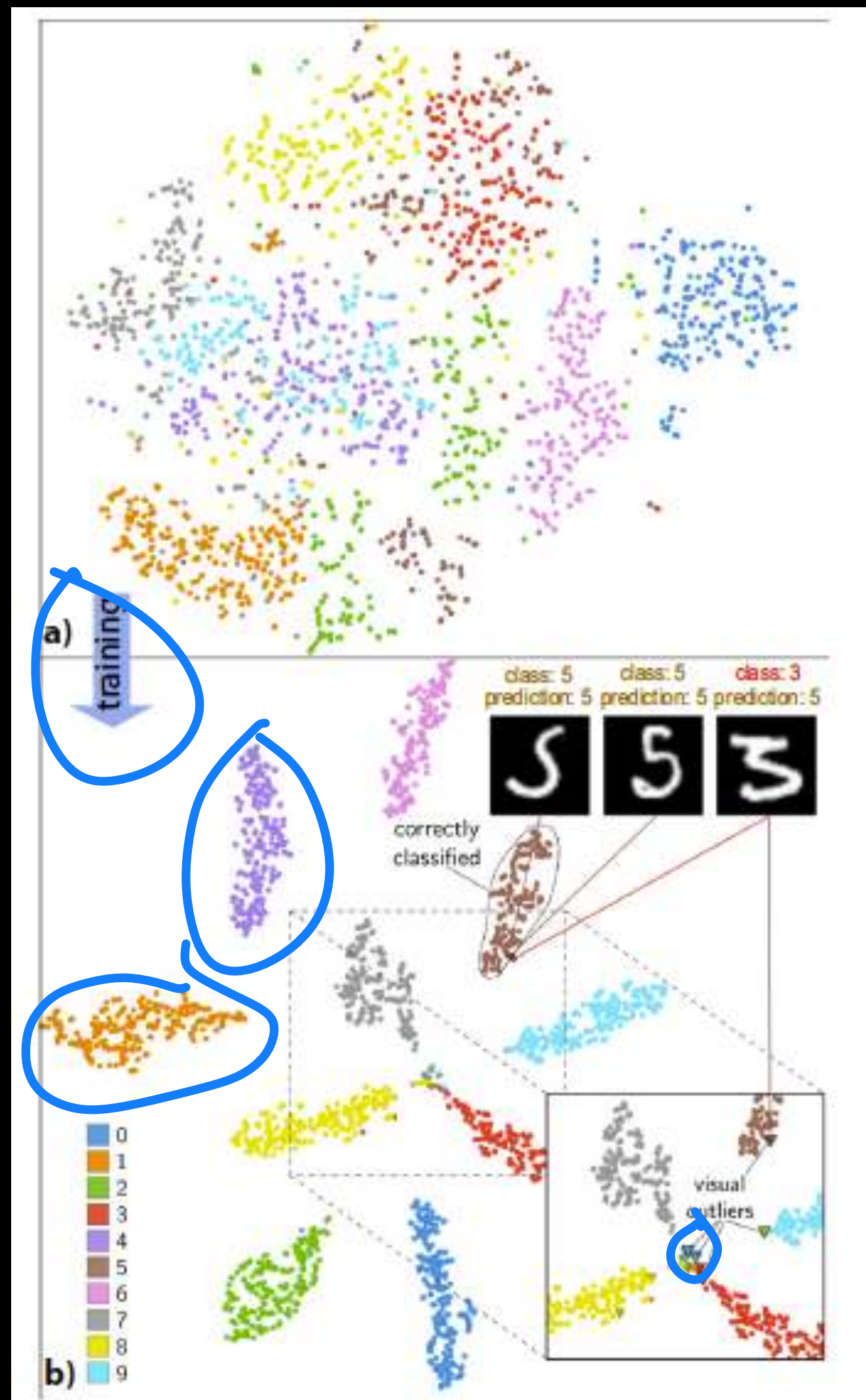
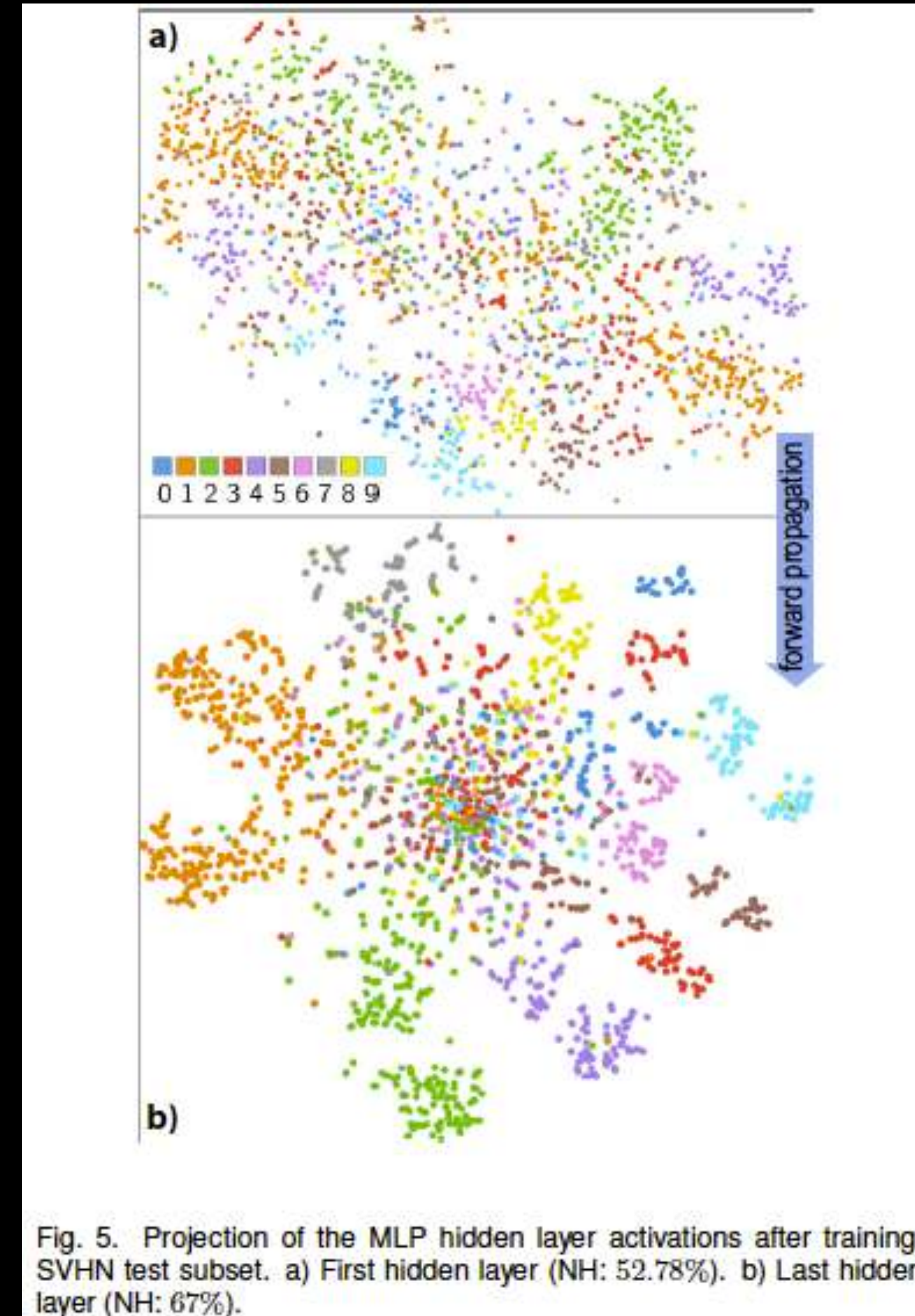
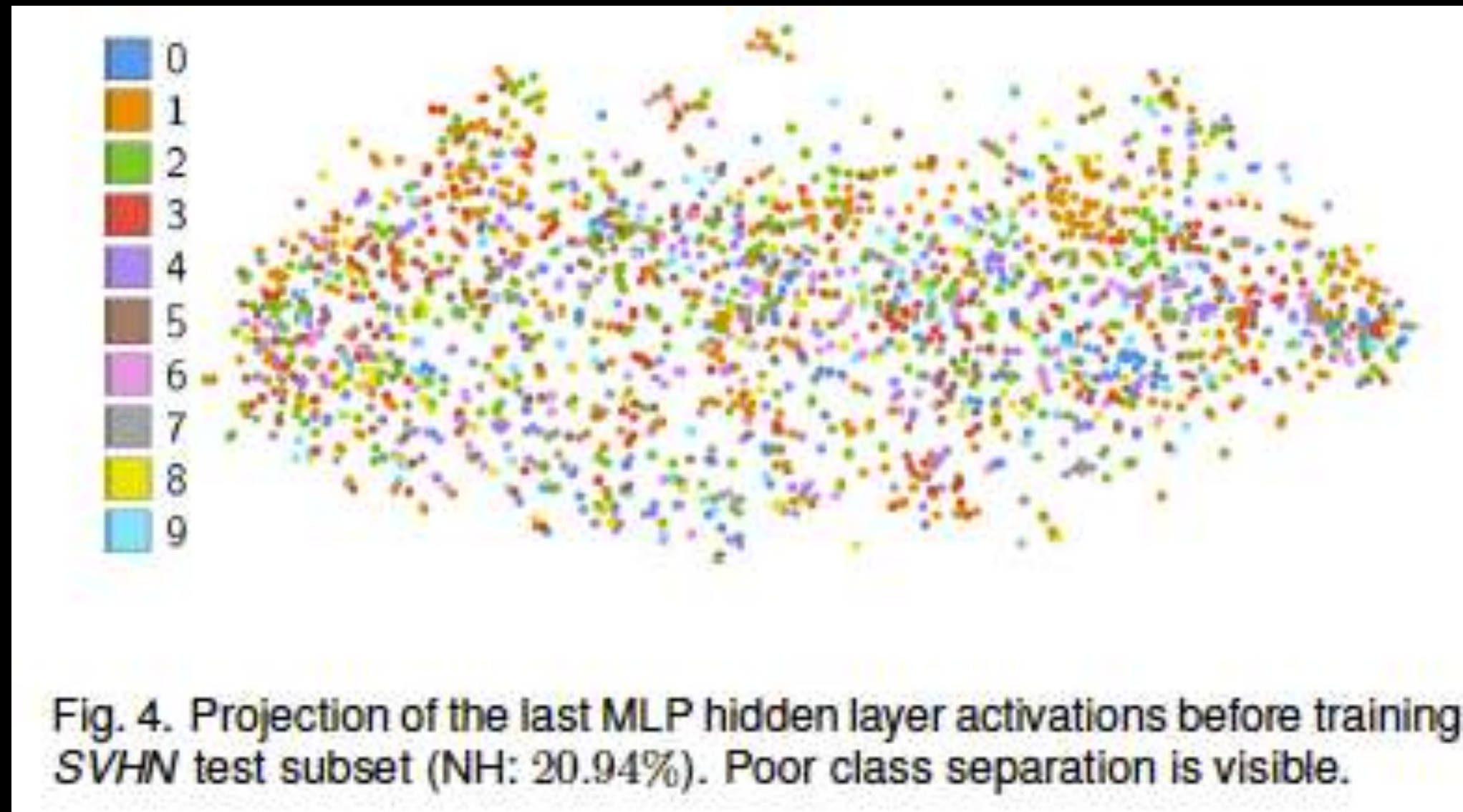
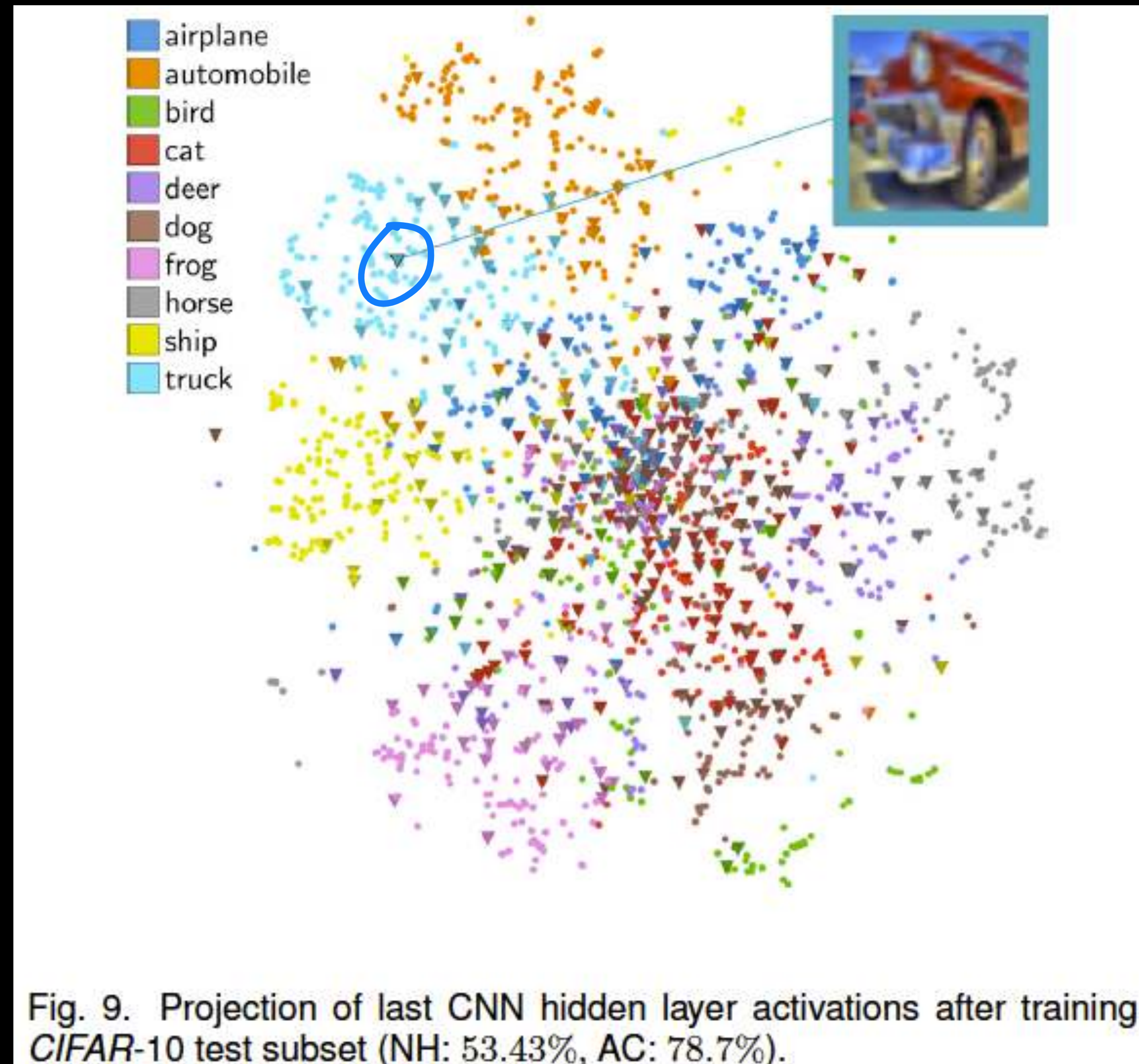


Fig. 3. Projection of the last MLP hidden layer activations, MNIST test subset. a) Before training (NH: 83.78%). b) After training (NH: 98.36%, AC: 99.15%). Inset shows classification of visual outliers.

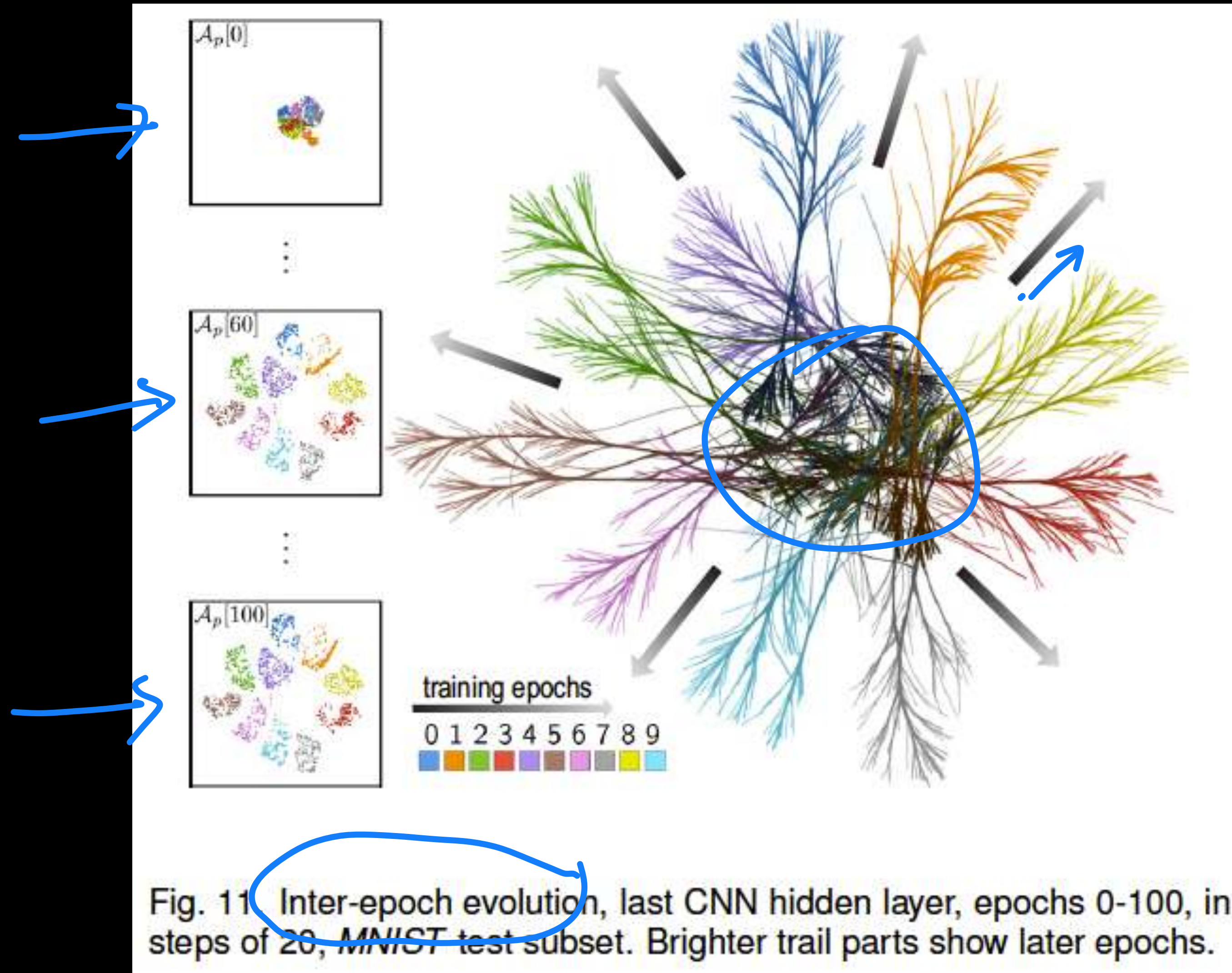
Understanding Deep Networks



Understanding deep networks



Understanding Deep Networks



Understanding Deep Networks

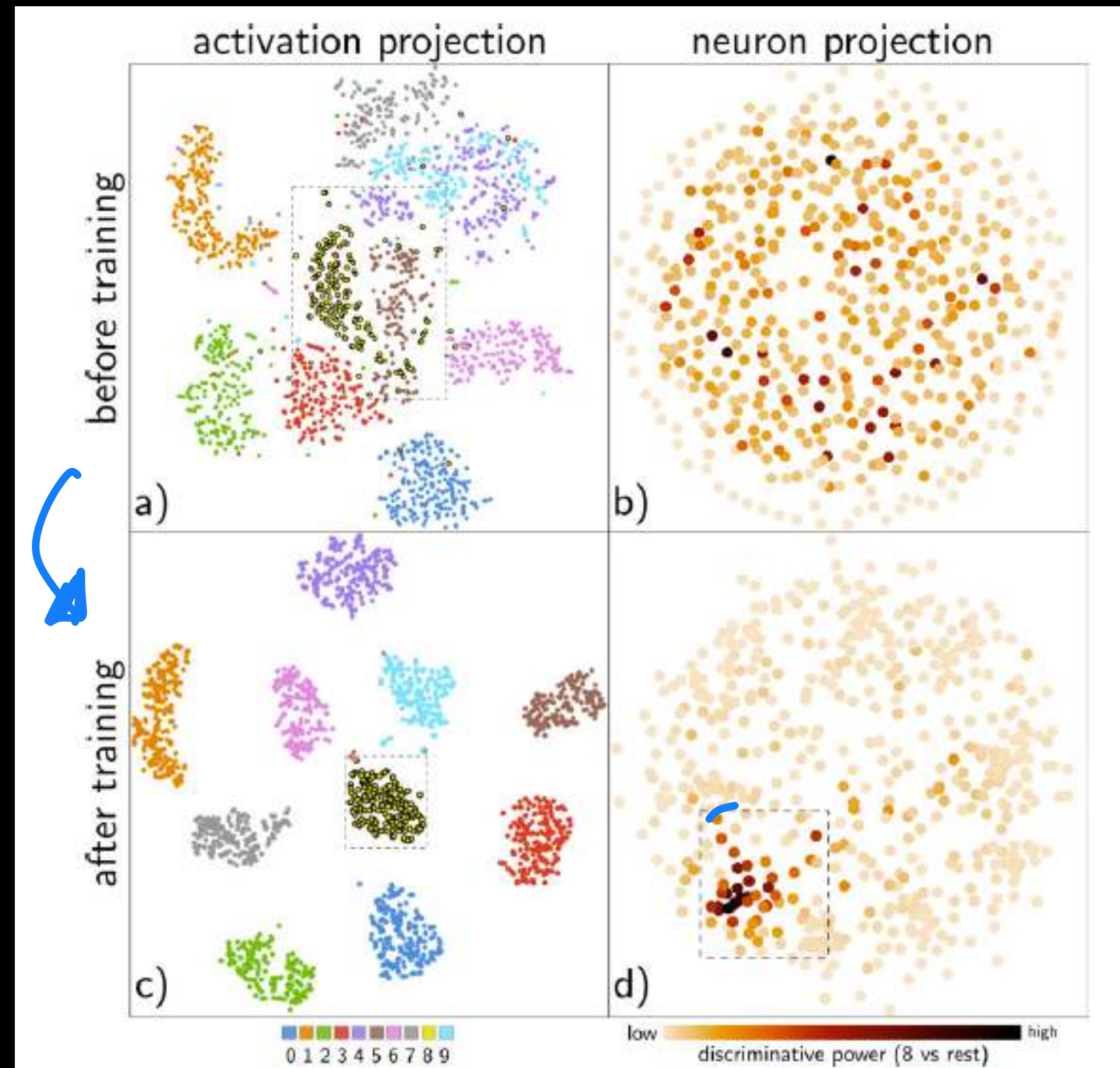
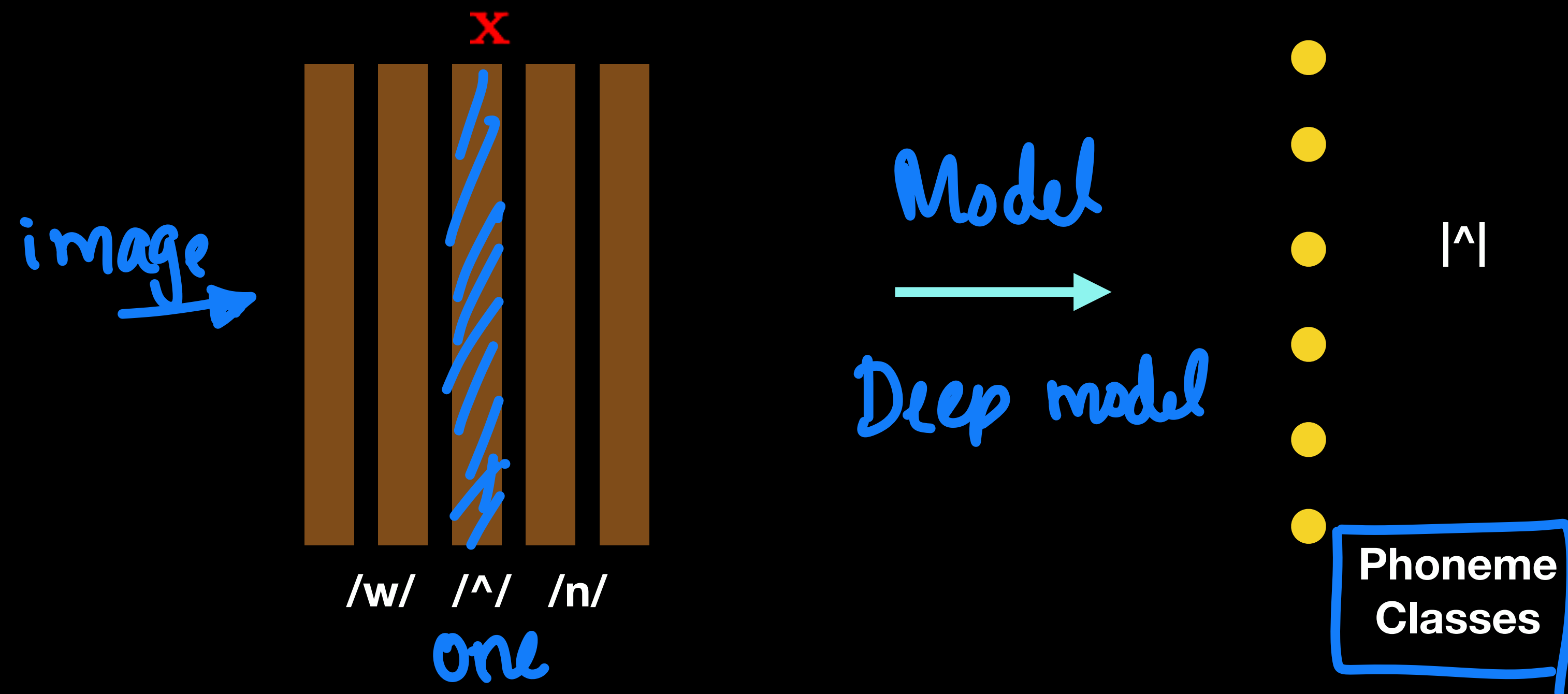


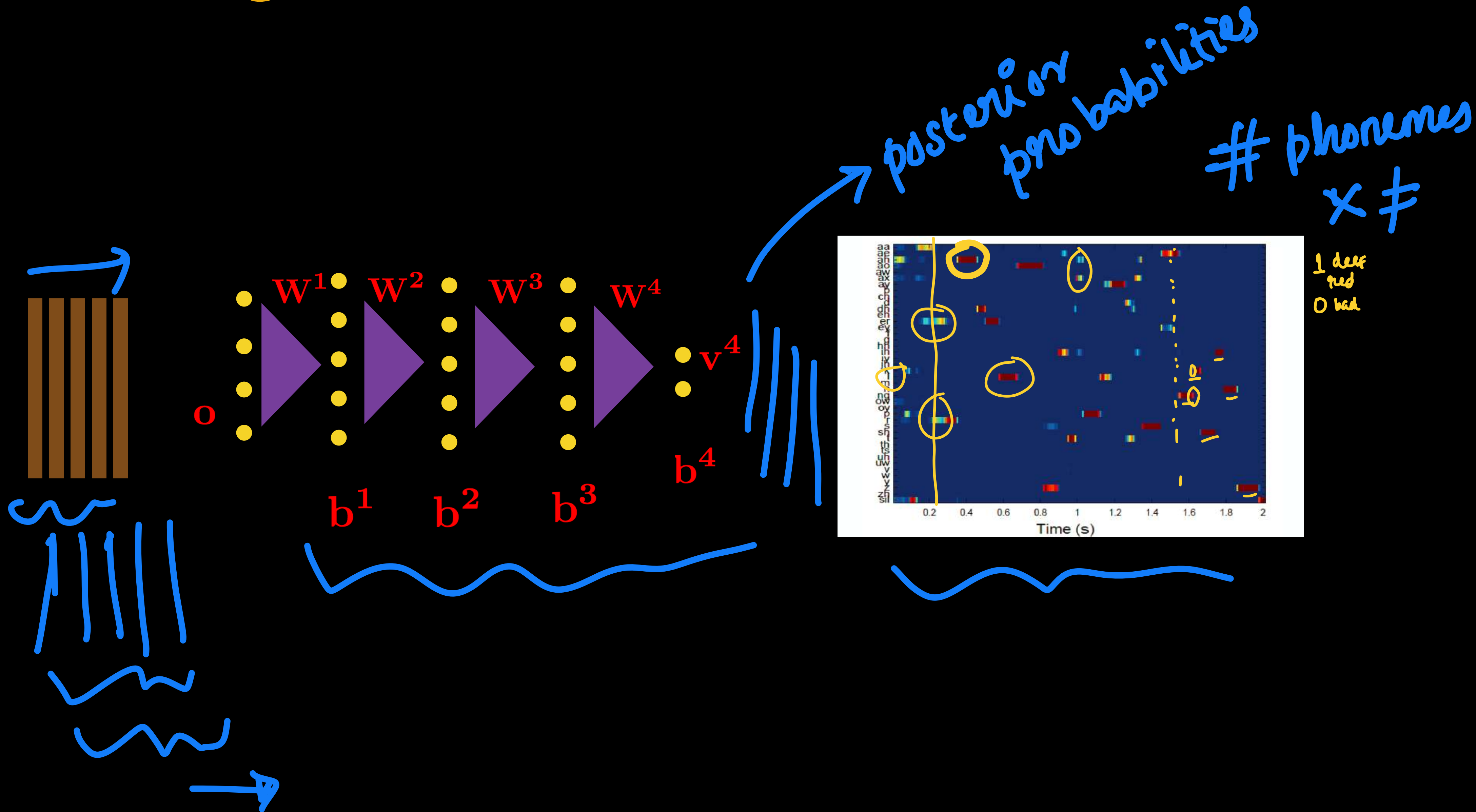
Fig. 12. Activation and neuron projections of last CNN hidden layer activations before and after training, *MNIST* test subset. Neuron projection colors show the neurons' power to discriminate class 8 vs rest.

Speech Recognition (Acoustic modeling)



- Classical machine learning - train a classifier on speech training data that maps to the target phoneme class.

Speech recognition



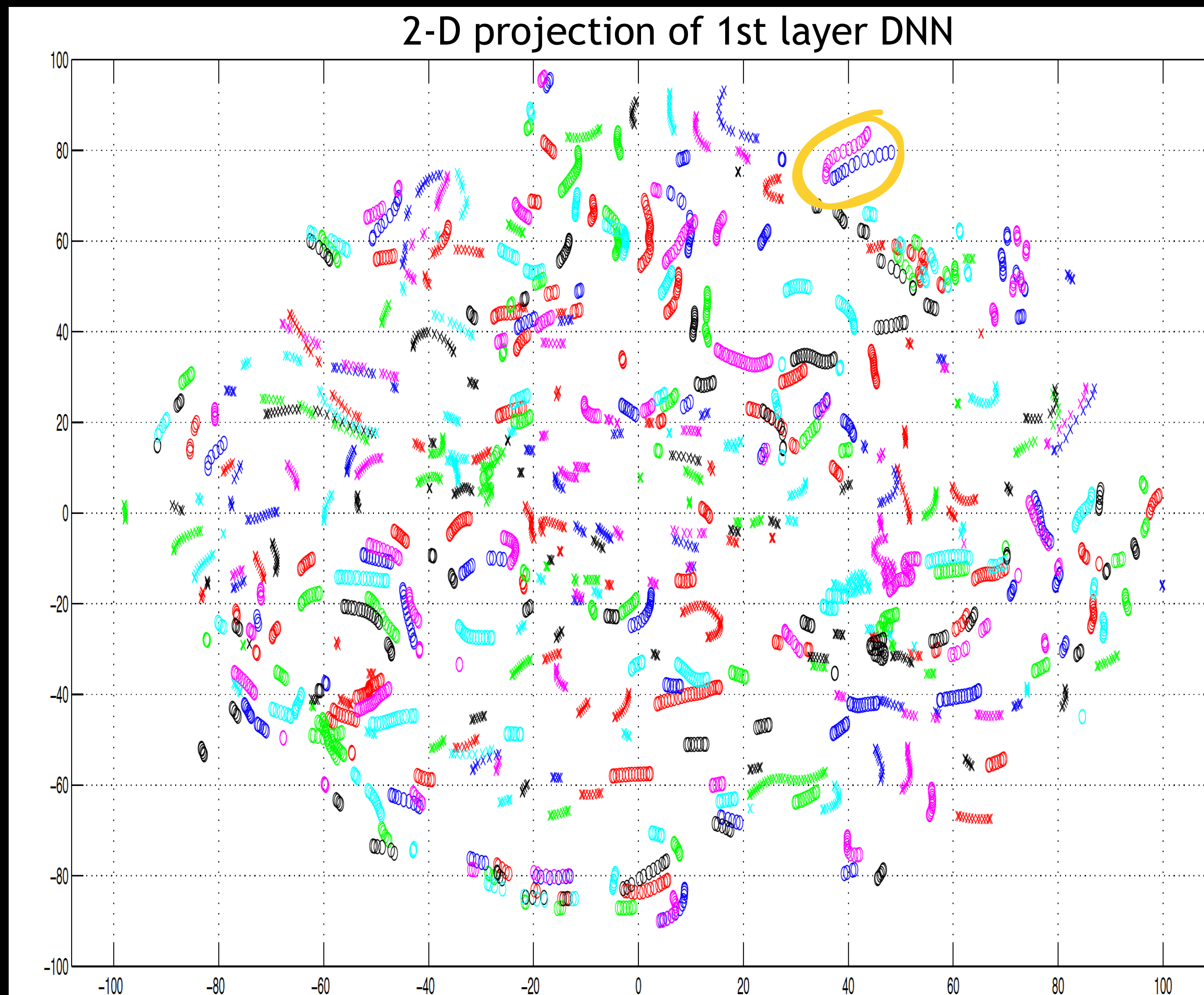
Understanding DNNs for speech

Mohamed, Abdel-rahman, Geoffrey Hinton, and Gerald Penn. "Understanding how deep belief networks perform acoustic modelling." 2012 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2012.

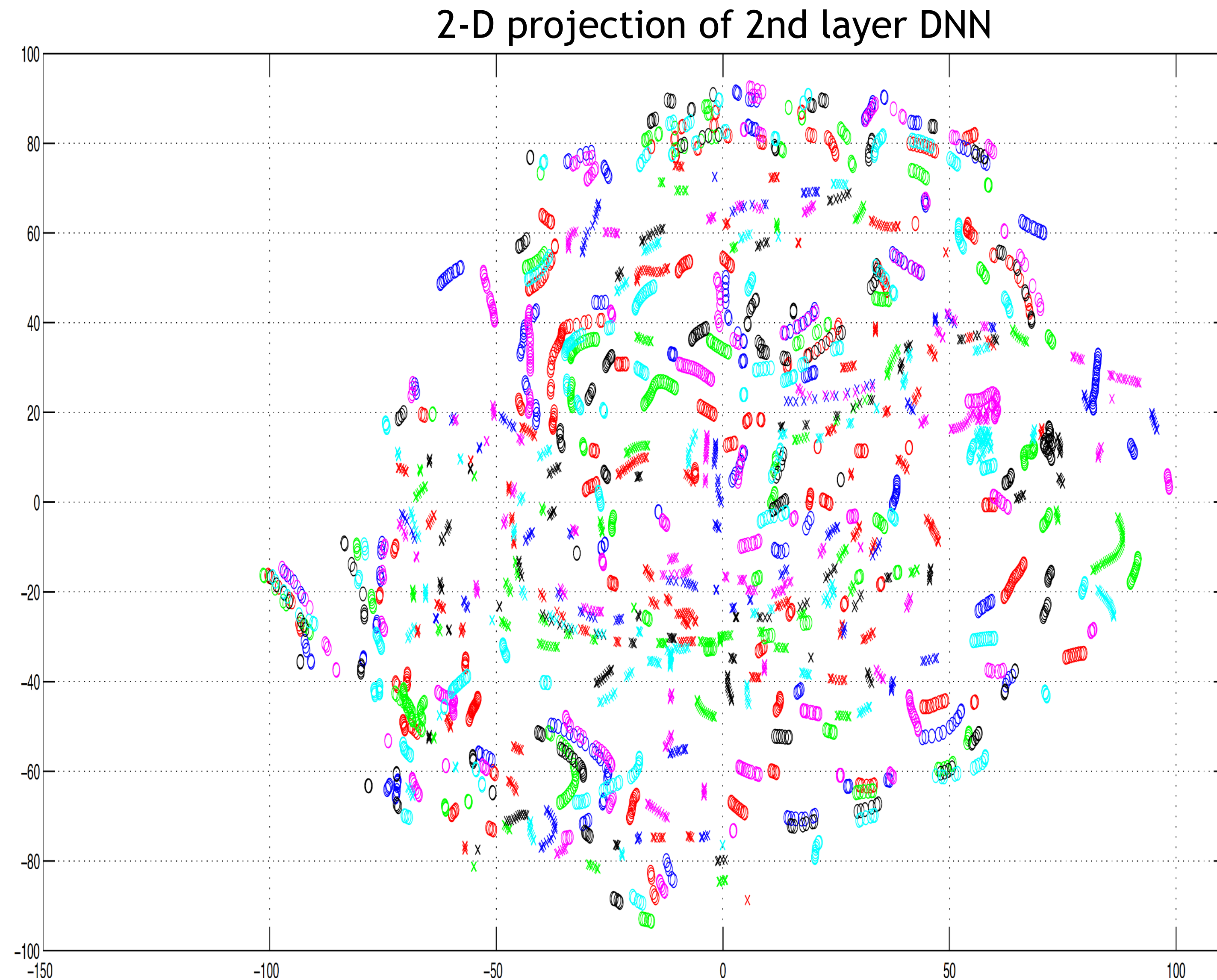
Understanding DNNs for Speech

[Abdel Rahman, 2012]

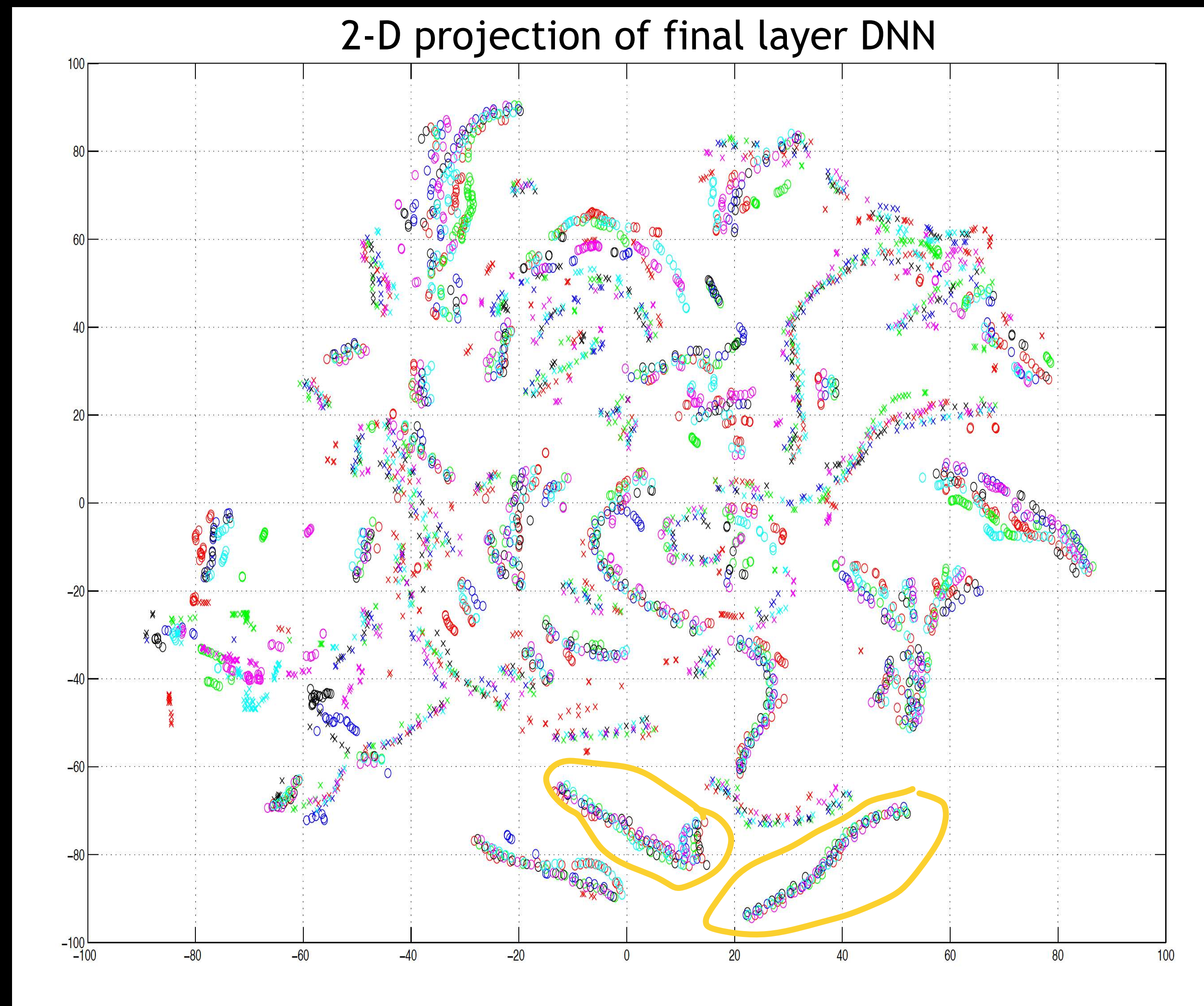
t-SNE
8-layer
DNN



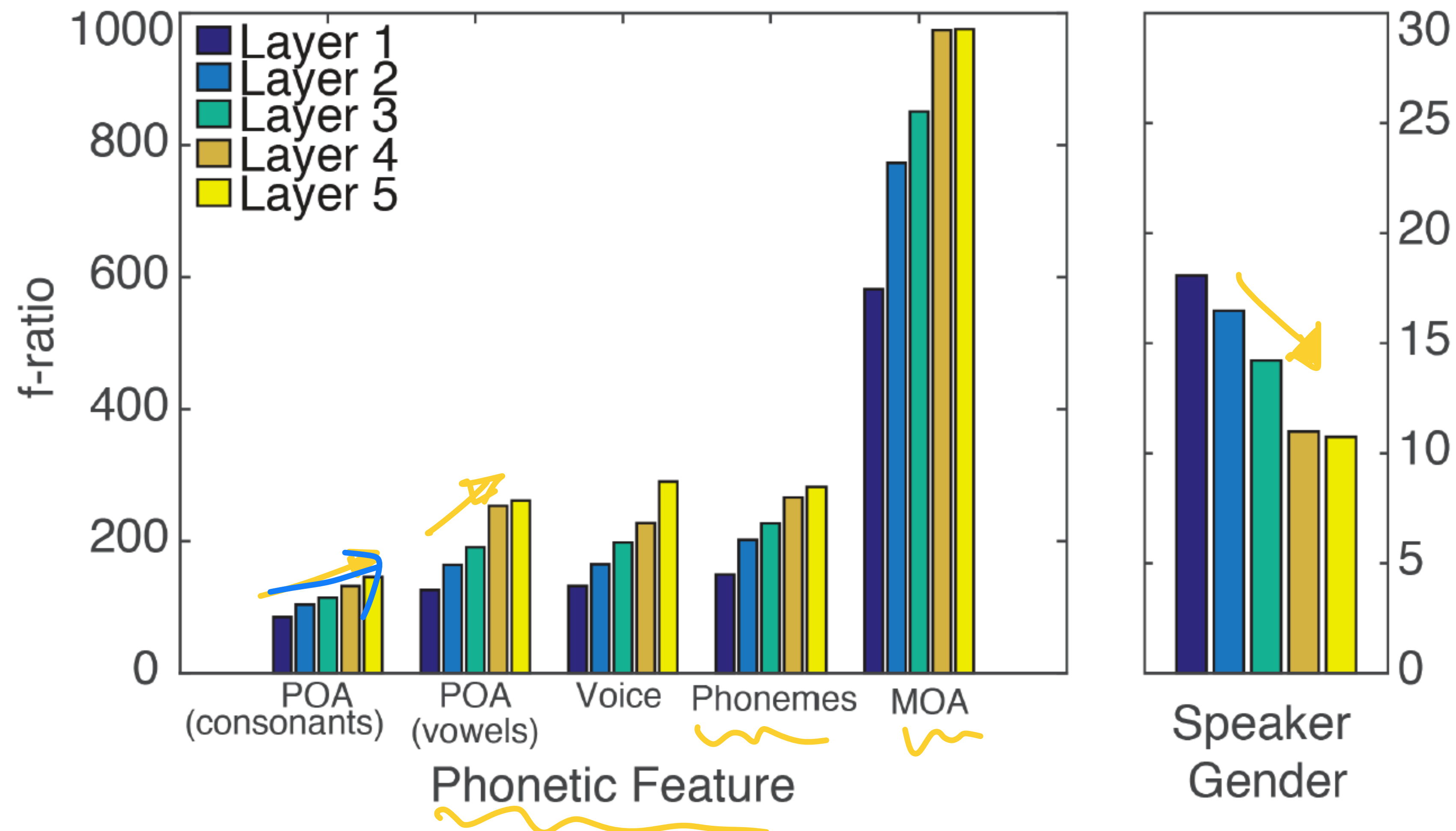
Understanding DNNs for Speech



Understanding DNNs for Speech



Understanding DNNs for Speech



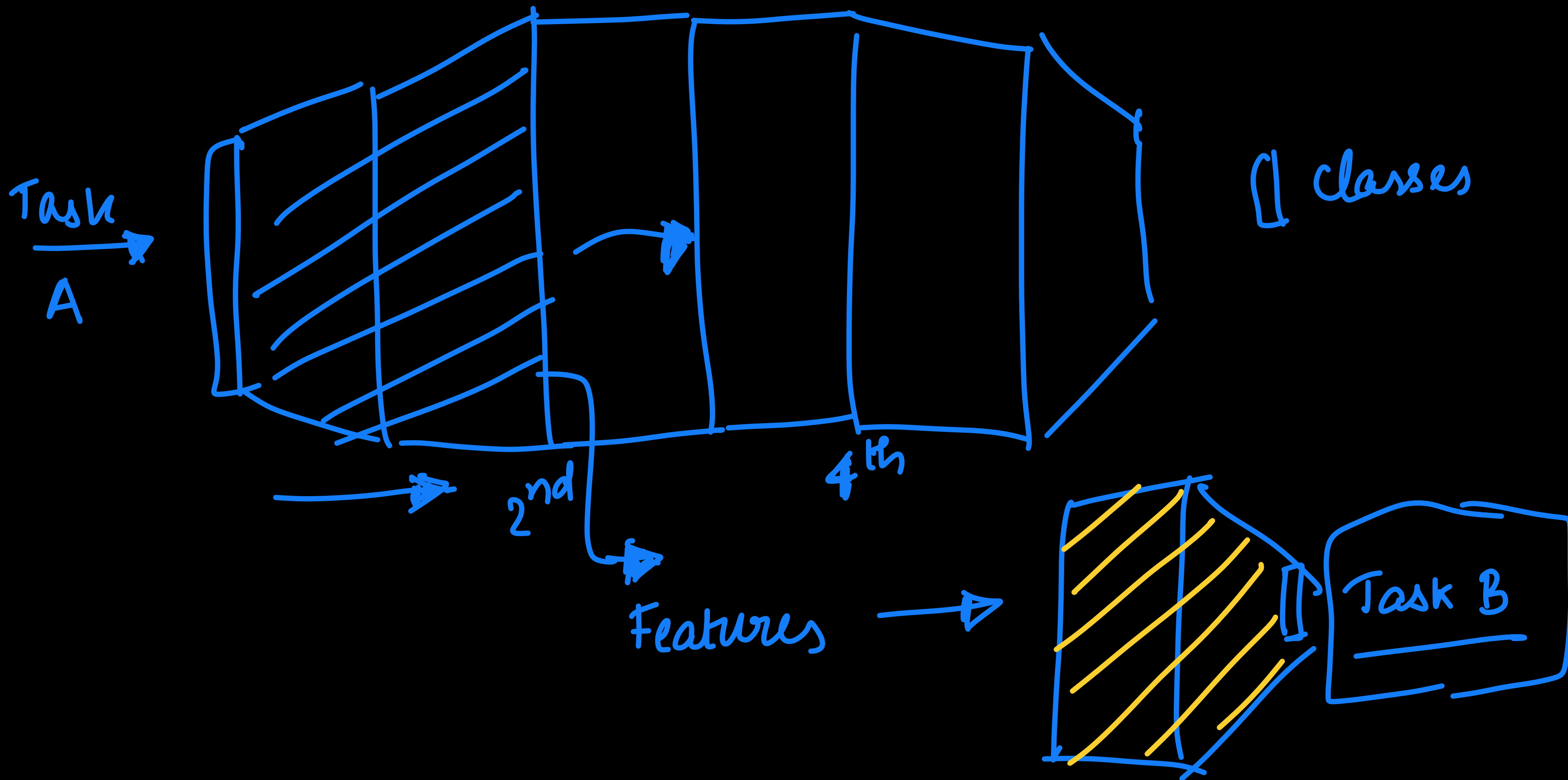
Summary thus far

- ★ Deep neural networks perform hierarchical data abstractions
 - ✓ Early layers form representations that are less oriented towards the task.
 - ✓ Later layers form representations that more oriented to the task.
- ★ Connections with biological processing of audio/images.

Questions about representations

- ✳ Can we quantify the degree to which a particular layer is general or specific?
- ✳ Does the transition occur suddenly at a single layer, or is it spread out over several layers?
- ✳ Where does this transition take place: near the first, middle, or last layer of the network?





Questions about representations

How transferable are features in deep neural networks?

2013

Jason Yosinski,¹ Jeff Clune,² Yoshua Bengio,³ and Hod Lipson⁴

¹ Dept. Computer Science, Cornell University

² Dept. Computer Science, University of Wyoming

³ Dept. Computer Science & Operations Research, University of Montreal

⁴ Dept. Mechanical & Aerospace Engineering, Cornell University



Questions about representations

A 500

B 500

pizza

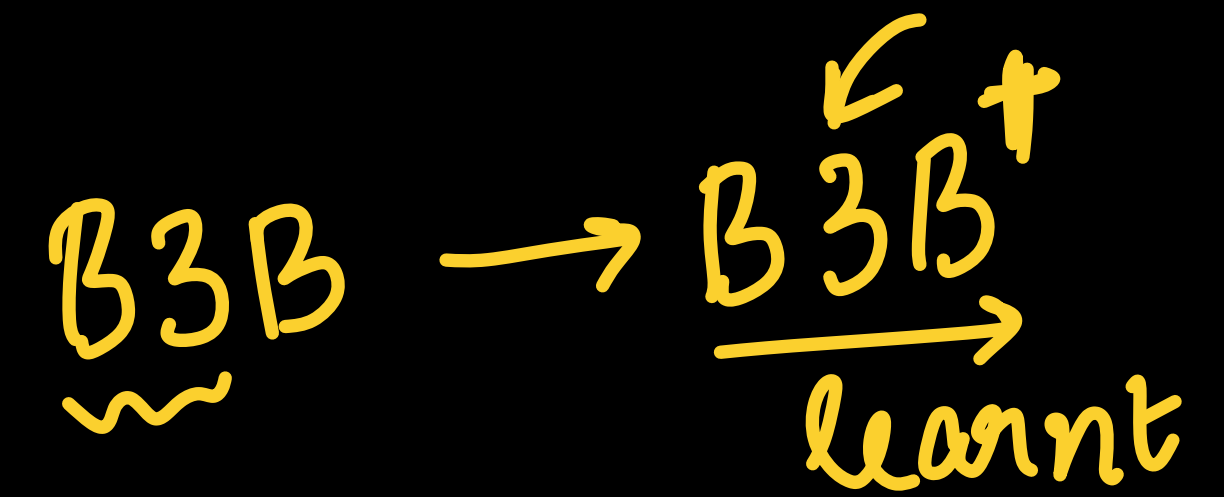
beer

Imagenet
Dataset

1000 images
1000 classes



Questions about representations



- ✳ A selfer network B3B: the first 3 layers are copied from baseB and frozen. The five higher layers (4–8) are initialized randomly and trained on dataset B. This network is a control for the next transfer network.
- ✳ A transfer network A3B: the first 3 layers are copied from baseA and frozen. The five higher layers (4–8) are initialized randomly and trained toward dataset B. Intuitively, here we copy the first 3 layers from a network trained on dataset A and then learn higher layer features on top of them to classify a new target dataset B.



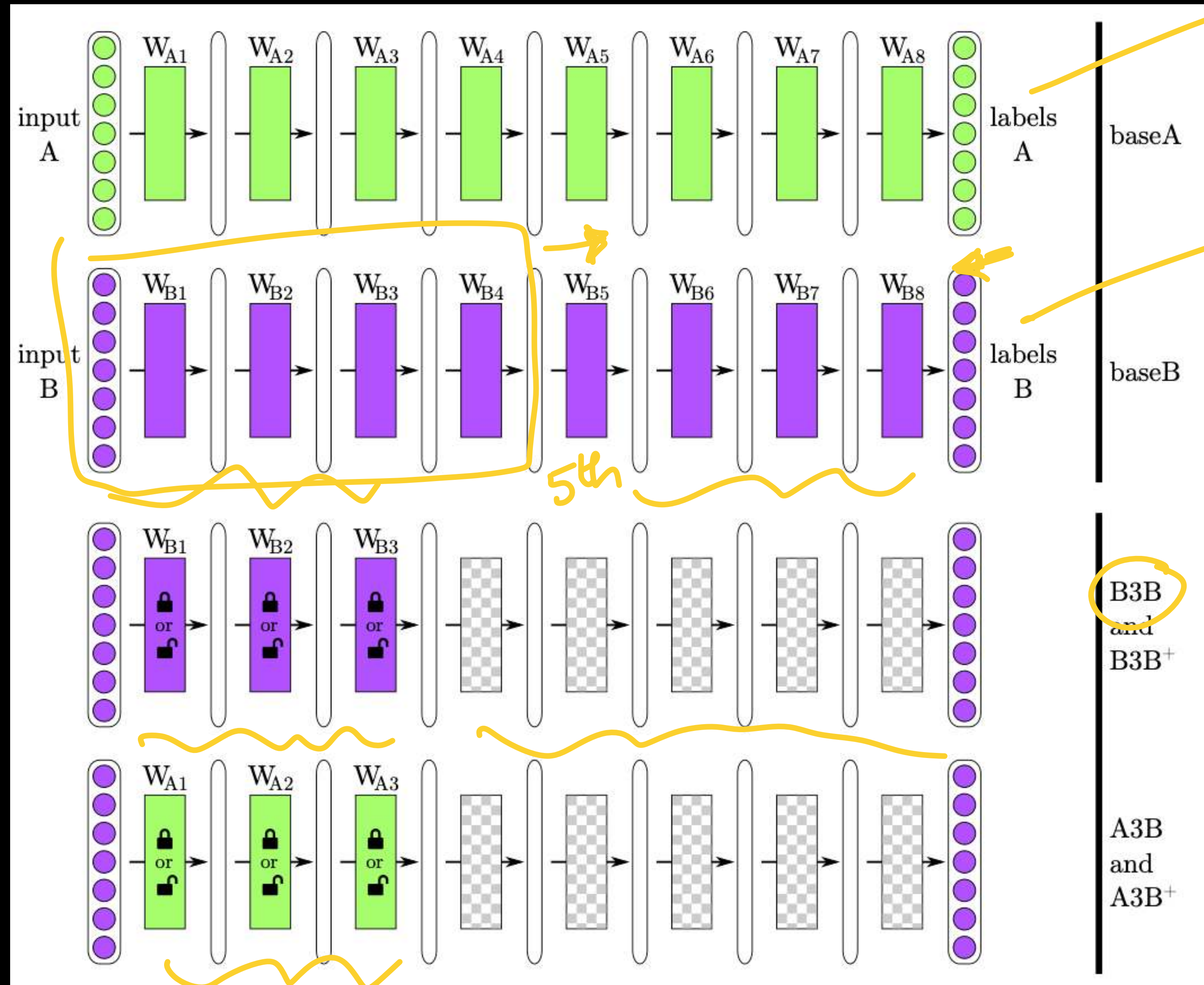
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Questions about representations

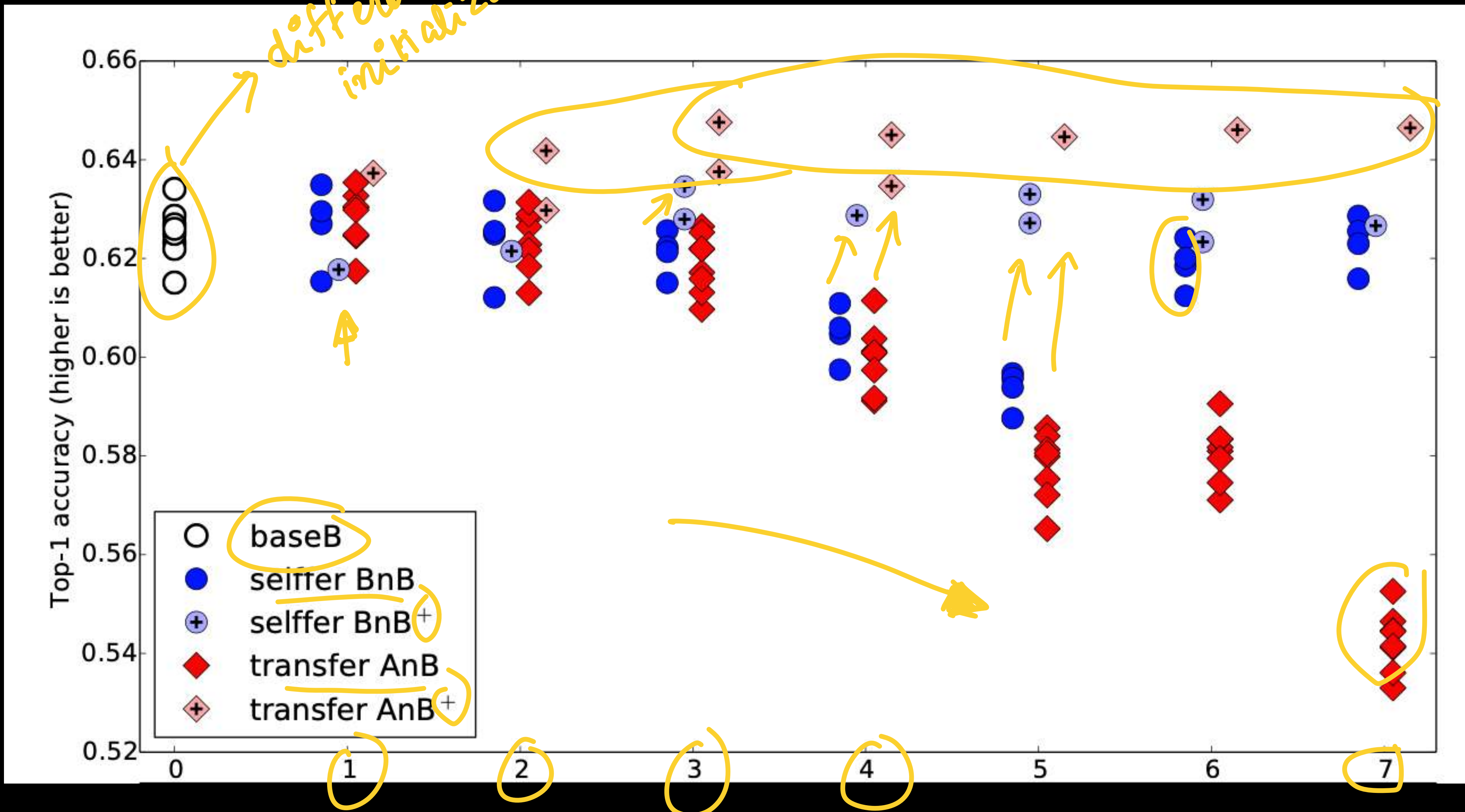
DCN



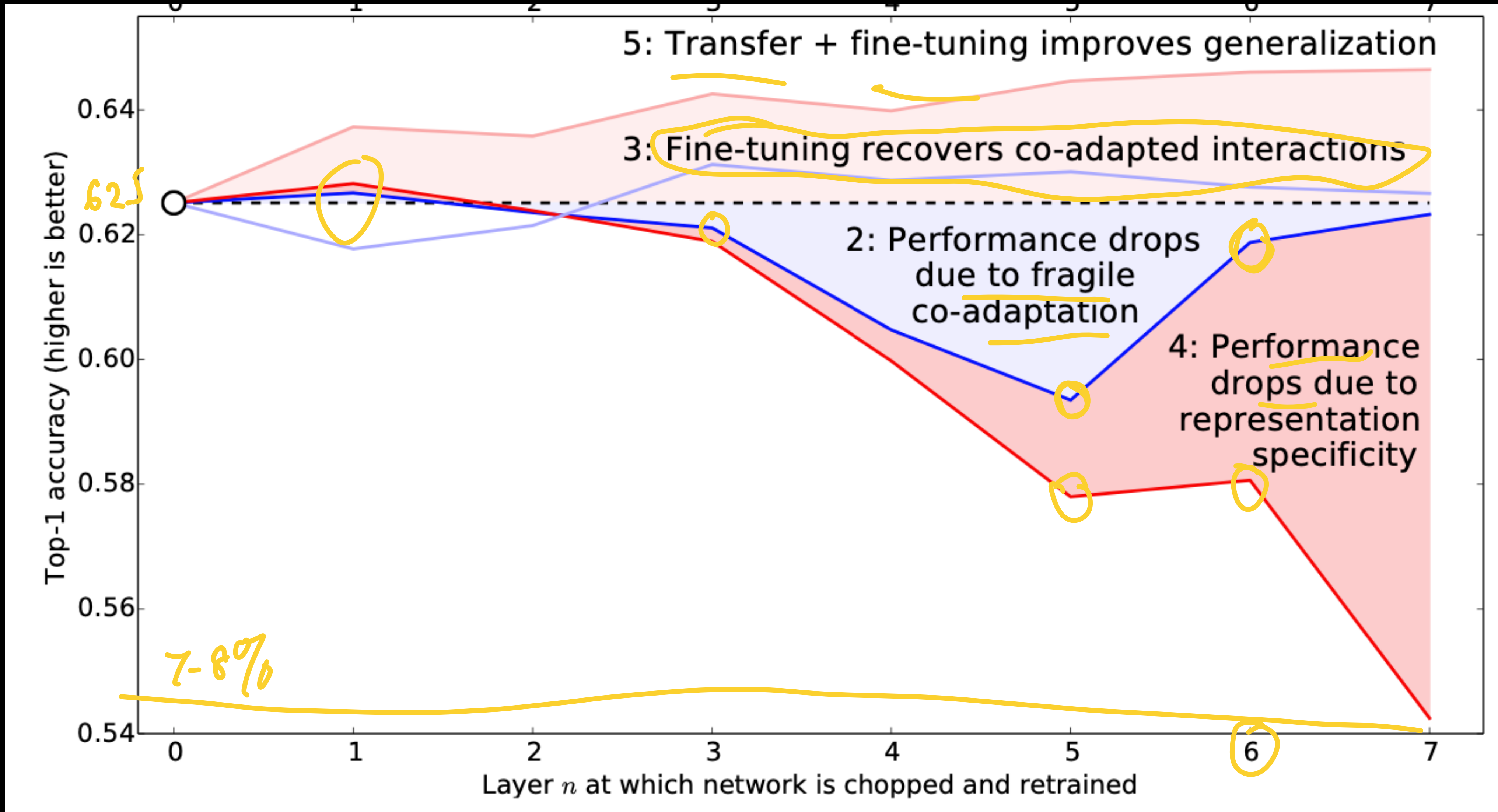
B3B
B3B → B3B⁺
A3B → A3B⁺



Questions about representations



Questions about representations



Questions about representations

living - 550
non-living - 450

